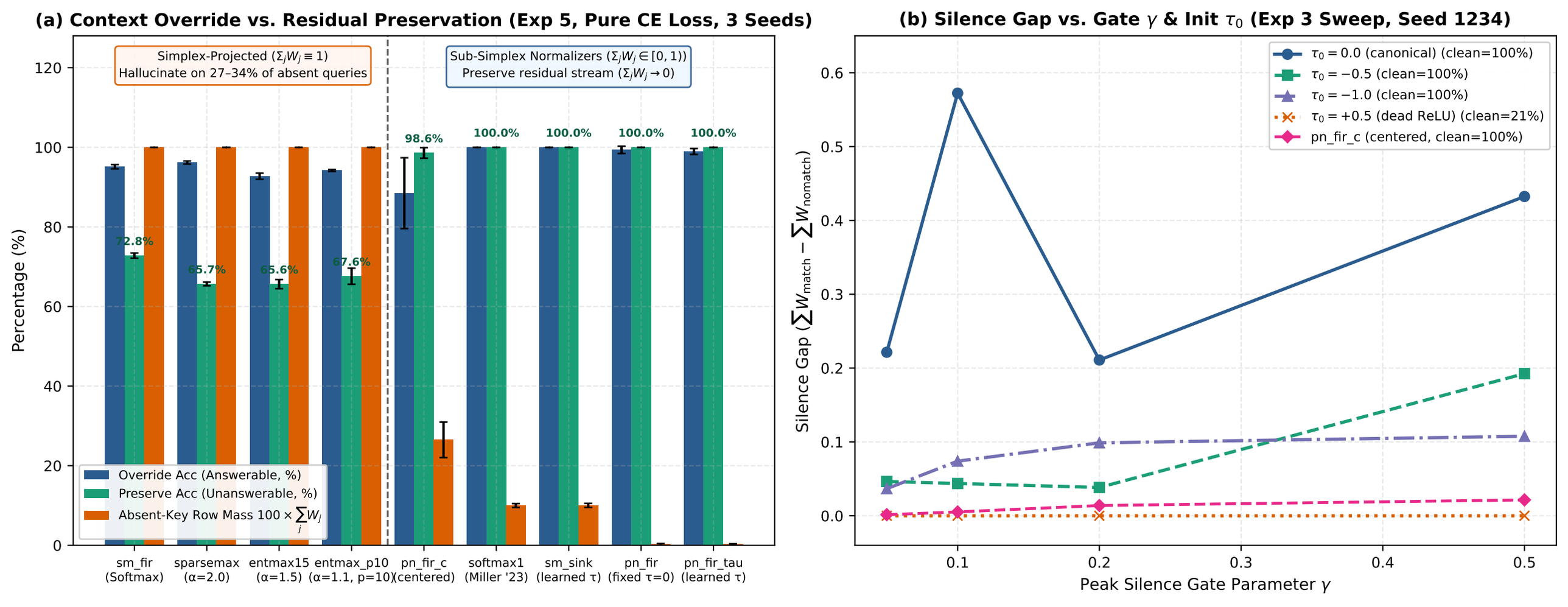
(a) Context Override vs. Residual Preservation (Exp 5, Pure CE Loss, 3 Seeds)
Simplex-Projected ($\Sigma_j W_j \equiv 1$)
Hallucinate on 27–34% of absent queries
Sub-Simplex Normalizers ($\Sigma_j W_j \in [0, 1)$)
Preserve residual stream ($\Sigma_j W_j \to 0$)
Percentage (%)
72.8%
65.7%
65.6%
67.6%
98.6%
100.0%
100.0%
100.0%
100.0%
Override Acc (Answerable, %)
Preserve Acc (Unanswerable, %)
Absent-Key Row Mass $100 \times \sum_j W_j$
sm_fir (Softmax)
sparsemax (α=2.0)
entmax15 (α=1.5)
entmax_p10 (α=1.1, p=10)
pn_fir_c (centered)
softmax1 (Miller '23)
sm_sink (learned τ)
pn_fir (fixed τ=0)
pn_fir_tau (learned τ)
(b) Silence Gap vs. Gate γ & Init τ_0 (Exp 3 Sweep, Seed 1234)
$\tau_0 = 0.0$ (canonical) (clean=100%)
$\tau_0 = -0.5$ (clean=100%)
$\tau_0 = -1.0$ (clean=100%)
$\tau_0 = +0.5$ (dead ReLU) (clean=21%)
pn_fir_c (centered, clean=100%)
Silence Gap ($\sum W_{match} - \sum W_{nomatch}$)
Peak Silence Gate Parameter γ